

Lecture 0 – Course Overview

AAA551: Programming Language Theory

Jihyeok Park



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- **Instructor:** Jihyeok Park (박지혁)
 - **Position:** Assistant Professor in CS, Korea University
 - **Expertise:** Programming Languages, Software Analysis
 - **Office hours:** 14:00–16:00, Tuesdays (appointment by e-mail)
 - **Office:** 507, Jung Woonoh IT & General Education Center
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- **Class:** AAA551: Programming Language Theory
- **Homepage:** <https://plrg.korea.ac.kr/courses/aaa551/>
- **LMS:** <https://lms.korea.ac.kr/>
- **Prerequisites:** Basic knowledge of programming languages and type systems (e.g., from COSE212 or equivalent courses) is expected.

- **2 Assignments: 80%**
 - Programming assignments in Scala (submission in [LMS](#))
- **Attendance: 20%**
 - Please use [LMS](#) to attend the class with the code provided.
 - No attendance check for the first week, midterm, and final week.
- There will be no offline lecture on the following day. Instead, recorded lecture videos will be uploaded to [LMS](#).
 - May 5 (Tue.) – National Holiday (어린이날)

- **Self-contained lecture notes.**

<https://plrg.korea.ac.kr/courses/aaa551/>

- **Reference:** You can refer to the following textbooks for more details on the topics covered in this course.
 - “Types and Programming Languages” by Benjamin C. Pierce.
 - “Advanced Topics in Types and Programming Languages” by Benjamin C. Pierce.
 - “Introduction to Static Analysis: An Abstract Interpretation Perspective” by Xavier Rival and Kwangkeun Yi.

- JavaScript
 - `[] + []` produces `""`
 - `[] + {}` produces `"[object Object]"`
 - `[] - []` produces `0`
 - `[] - {}` produces `NaN`
- Python

```
funcs = []  
  
for k in range(3):  
    funcs.append(lambda x: x + k)  
  
print(funcs[0](10))  
print(funcs[1](10))  
print(funcs[2](10))
```

It prints 12 three times, not 10, 11, and 12.

- Scala

```
trait Parent:  
  val x: Int  
  val y = x * 2  
  
object Child extends Parent:  
  val x = 10  
  
println(Child.y)
```

It prints 0, not 20.

- We need to replace fuzzy intuition with **mathematical precision**.
- To bridge the gap between “*what we think code does*” and “*what it actually does*”, we need **formal semantics**.

What is Semantics?

- Semantics is simply a fancy way of saying “Meaning”.
- Goal: To give a precise meaning to programs.

Why is this useful?

- For Programmers:
 - To clearly understand what our code is going to do before we run it.
- For Tool Builders (Compilers, Optimizers):
 - How do we know an optimization is safe?
 - Semantics allows us to judge whether these tools are implemented correctly.

A Deeper Question: “What is Information?”

- Standard Information:
 - Familiar mathematical tools like sets and sequences are adequate for static data.
- Programs as Information:
 - Programs describe computation, but they are also a kind of information.
 - They are a particularly interesting, dynamic kind of information.
- The Challenge:
 - Standard math is not enough.
 - We need specialized formal tools to precisely describe this unique type of information.

We view Programming Languages through two complementary lenses:

- **Semantics (The Meaning)**
 - **Operational:** *How* is a program executed?
 - **Denotational:** *What* is the mathematical object for a program?
 - **Axiomatic:** *Which properties* does a program satisfy?
- **Type Systems (The Guardrails)**
 - **Foundations:** Simply-Typed Lambda Calculus, Extensions
 - **Type Abstraction:** Universal, Subtyping, Existential, Operators
 - **Propositions as Types:** Curry-Howard, Dependent Types
 - **Modern Types:** Refinement, Flow-Sensitive, Substructural, Effects

- Basic Introduction to Scala

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